

SE3062 – Intelligent Systems

Lecture 02 – Intelligent Agents

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Recap From Last Week

Why are Humans Intelligent?

- Because of the exceptional Learning Process.
- Key Aspects of Human Learning:
 - **Sensory Perception:** Humans gather information from their surroundings using sensory organs.
 - **Memory and Knowledge Storage:** Past experiences and knowledge are stored for future reference.
 - **Decision-Making:** Humans analyze available information and make decisions based on reasoning and intuition.
 - **Adaptation and Learning:** Continuous improvement occurs through learning from successes and mistakes.

Different View points

Thinking Humanly “The exciting new effort to make computers think . . . <i>machines with minds</i> , in the full and literal sense.” (Haugeland, 1985) “[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . .” (Bellman, 1978)	Thinking Rationally “The study of mental faculties through the use of computational models.” (Charniak and McDermott, 1985) “The study of the computations that make it possible to perceive, reason, and act.” (Winston, 1992)
Acting Humanly “The art of creating machines that perform functions that require intelligence when performed by people.” (Kurzweil, 1990) “The study of how to make computers do things at which, at the moment, people are better.” (Rich and Knight, 1991)	Acting Rationally “Computational Intelligence is the study of the design of intelligent agents.” (Poole <i>et al.</i> , 1998) “AI . . . is concerned with intelligent behavior in artifacts.” (Nilsson, 1998)

Intelligent Agents

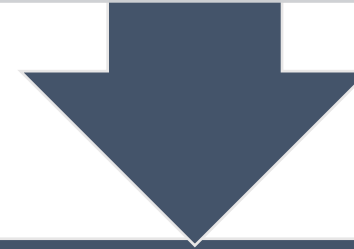
From Intelligence to Agency

Intelligence is the ability to:

Perceive the
environment

Reason and
learn

Take actions to
achieve goals



In AI, this translates to **building agents** that act **rationally** and **effectively** in their environments

From Intelligence to Agency

Can it think like a human? → Can it act effectively?

Acting rationally doesn't require replicating humans - only solving problems well.

What is an Intelligent Agent?

- A "human agent" refers to a person performing a task or interacting with a system, while an "intelligent agent" is a computer program designed to autonomously make decisions and take actions based on its environment, essentially mimicking human-like intelligence.



From Thinking Machines to Acting Agents

- Real-world AI must:
 - Operate autonomously
 - Handle uncertainty
 - Adapt to changing environments
 - Make timely decisions
- This calls for intelligent agents — not just rule-followers or logic engines.

Application	Agent Behavior
Self-driving car	Senses environment, plans path, reacts in real time
Virtual assistant	Understands commands, retrieves info, schedules tasks
Game AI	Adapts strategy based on player behavior



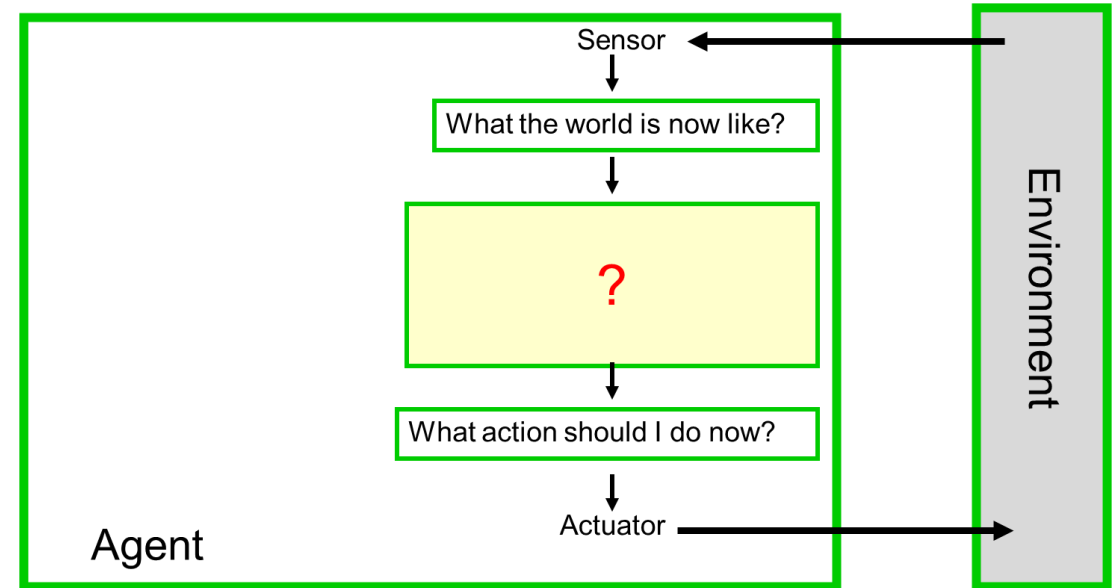
Question

- Which of your daily tools could be considered an intelligent agent?

What is an Agent?

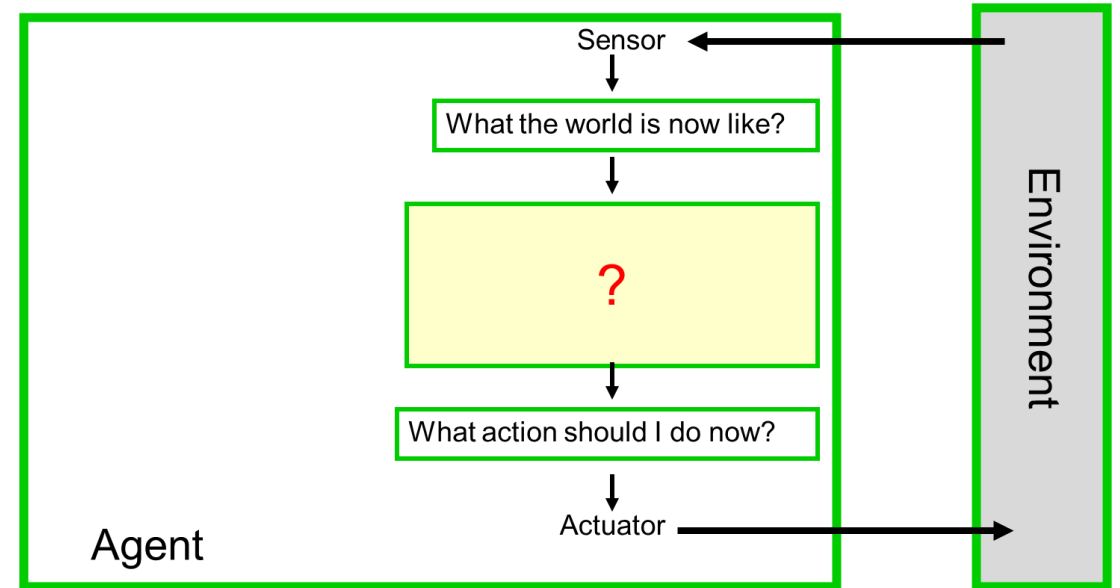
- Agent: Anything that perceives its environment through sensors and acts upon that environment through actuators.
- The agent operates in an environment.
- The AI system's intelligence depends on how well it perceives and acts.
- An agent is a function from percept histories to actions.

$$f: P^* \rightarrow A$$



A Vanilla Intelligent Agent

- **Sensors:**
 - cameras, touch sensor and infrared range finders
- **Actuator:**
 - various motors



What Makes an Agent Rational?

- A rational agent:
 - Chooses actions that maximize expected performance
 - Does the right thing, given what it knows
 - Can function under uncertainty
- A **rational agent** is not necessarily **intelligent**, but it **optimizes** decision-making based on its knowledge and goals.
 - Example: A **stock trading bot** makes **rational decisions** by analyzing market trends but does not possess **human-like intelligence**.
- Being rational $\neq$ being omniscient or always successful

Key Considerations:

- Rationality is based on:
 - **Percepts** received so far
 - **Knowledge** of the environment
 - **Available actions**
 - **Performance measure**

Environments and PEAS

- Why PEAS?
 - Helps us structure problems clearly
 - Necessary to design appropriate agent programs

COMPONENT

DESCRIPTION

P – Performance Measure

What defines success

E – Environment

What the agent operates in

A – Actuators

What the agent can do

S – Sensors

What the agent can perceive

Agent's Performance Measure



What should be a
**performance
measure** for a
robotic vacuum-
cleaner agent?

In-Class Group Activity

In small groups (2–3), pick one system:

Smart assistant, home security drone, chat-based tutor, etc.

Define its PEAS components

Share your example in 2 mins

Examples

Agent	P	E	A	S
Self-driving car	Safe, fast, legal driving	Roads, traffic, weather	Steering, throttle, brakes	Cameras, GPS, radar
Robotic vacuum	Clean floor, avoid collisions	Rooms, furniture, pets	Wheels, vacuum motor	IR sensors, cliff sensors
Stock trading bot	Max profit, low risk	Market prices, trends	Buy/sell/hold commands	Price feeds, financial news

Environment and Agent Types and Architectures

Environment types

Fully observable (vs. partially observable)

- An agent's sensors give it access to the complete state of the environment at each point in time.
- A **chess game** is fully observable (entire board is visible), but **autonomous driving** is partially observable (limited sensor range).

Deterministic (vs. stochastic)

- The next state of the environment is completely determined by the current state and the action executed by the agent
- Chess is **deterministic**, but poker is **stochastic** (depends on chance and other players).

Episodic (vs. sequential)

- The agent's experience is divided into atomic "episodes" (each episode consists of the agent perceiving and then performing a single action), and the choice of action in each episode depends only on the episode itself.

Environment types . . .

Static (vs. dynamic):

- The environment is unchanged while an agent is deliberating. (The environment is semi dynamic if the environment itself does not change with the passage of time but the agent's performance score does).
- A chessboard is static as it remains unchanged while the player decides a move, while traffic navigation is dynamic since other vehicles and pedestrians constantly change the environment.

Discrete (vs. continuous):

- A limited number of distinct, clearly defined percepts and actions.
- Board games are discrete since they have a finite set of moves, while robotic movement is continuous as it involves precise, fluid control.

Single agent (vs. multiagent):

- An agent operating by itself in an environment.
- A robot vacuum is a single-agent system, while a multiplayer online game is a multiagent system where different AIs or players interact.

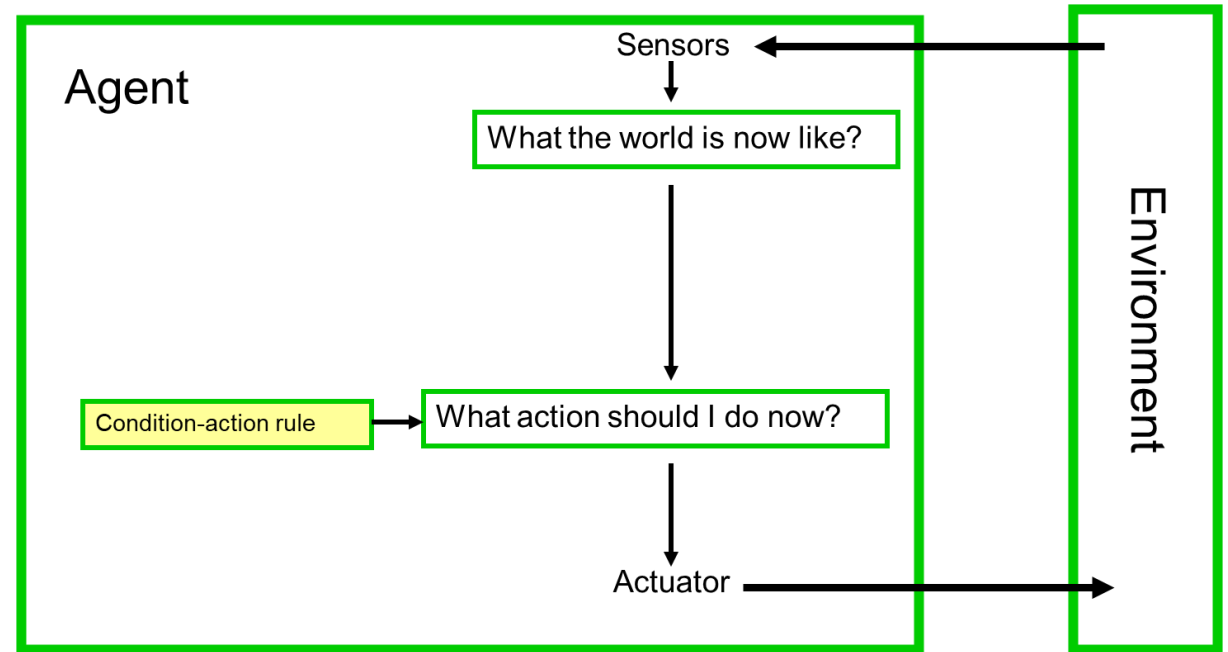
Agent types

1. Simple Reflex Agents
2. Model-Based Reflex Agents
3. Goal-Based Agents
4. Utility-Based Agents
5. Learning Agents

- **Each level adds more complexity and capability.**

Simple reflex agents

- Act only on current percept
- No memory or internal state
- Use condition–action rules
- Efficient but very limited
- **Example:**
If [bump detected] → then [turn left]



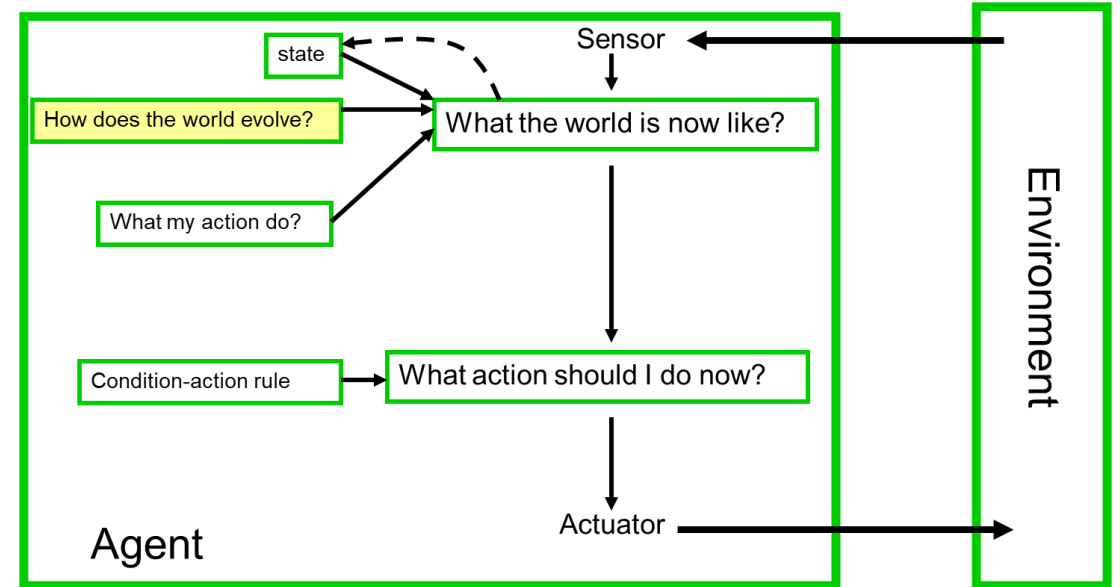
Agent Type	Description	Key Characteristics	Strengths	Limitations
Simple Reflex Agents	Act based on condition-action rules (if-then rules) without considering history.	- No memory of past states. - Works only in fully observable environments.	- Fast decision-making. - Simple and easy to implement.	- Cannot handle complex situations. - Fails in partially observable environments.

Simple
reflex
agents

- React based on current perceptions without memory of past states.
 - Example: A thermostat, which turns heating on or off based on the current temperature.

Model-based reflex agents

- Maintains internal model of the world.
- Can act even with partial information.
- More flexible than simple reflex agents.
- **Example:**
If [room is dirty] → clean
If [room was clean, now moved] → assume still clean



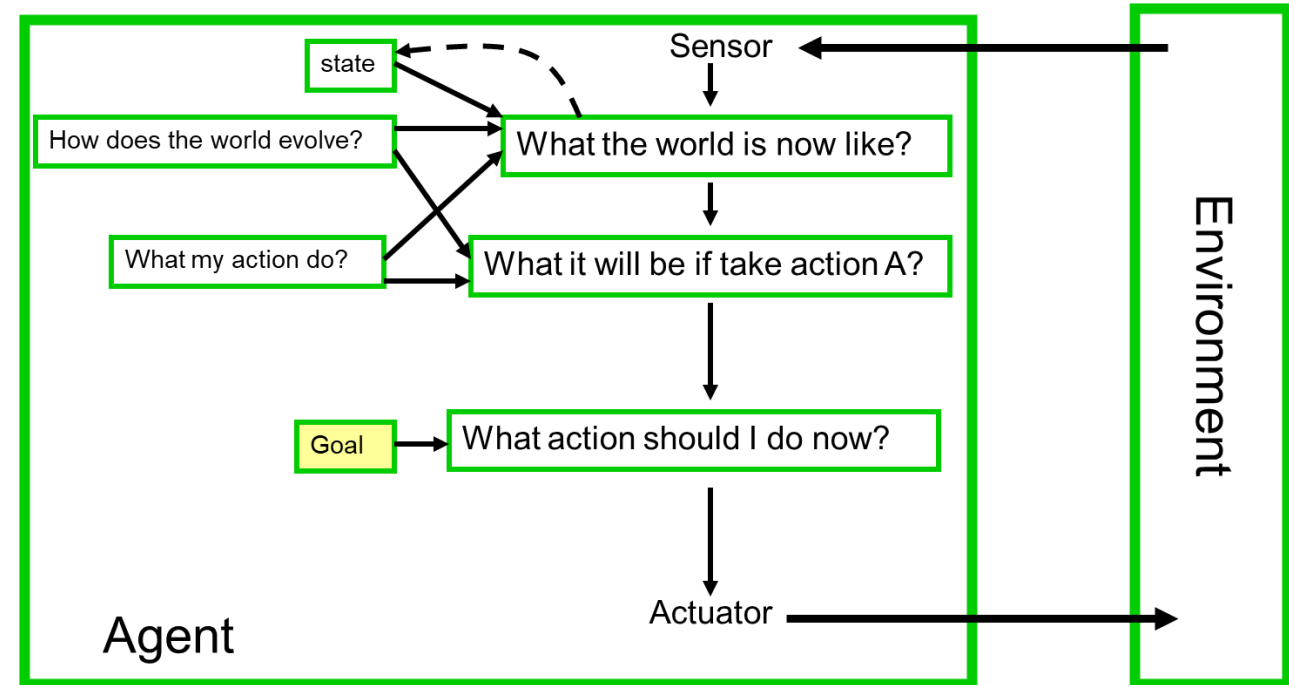
Agent Type	Description	Key Characteristics	Strengths	Limitations
Model-Based Reflex Agents	Use an internal model of the environment to make decisions, considering past states.	- Maintains internal state representation. - Can function in partially observable environments.	- More adaptable than simple reflex agents. - Can work with limited observations.	- More complex to design. - May require more computation.

Model-based reflex agents

- Maintain an internal model of the world to track past states and predict future changes.
 - Example: A self-driving car that remembers road signs and adjusts speed accordingly.

Goal-Based Agents

- Have explicit goals to pursue.
- Use search and planning.
- Choose actions that lead toward goal states.
- **Example:**
Goal: Reach charging station
→ Plan path around obstacles



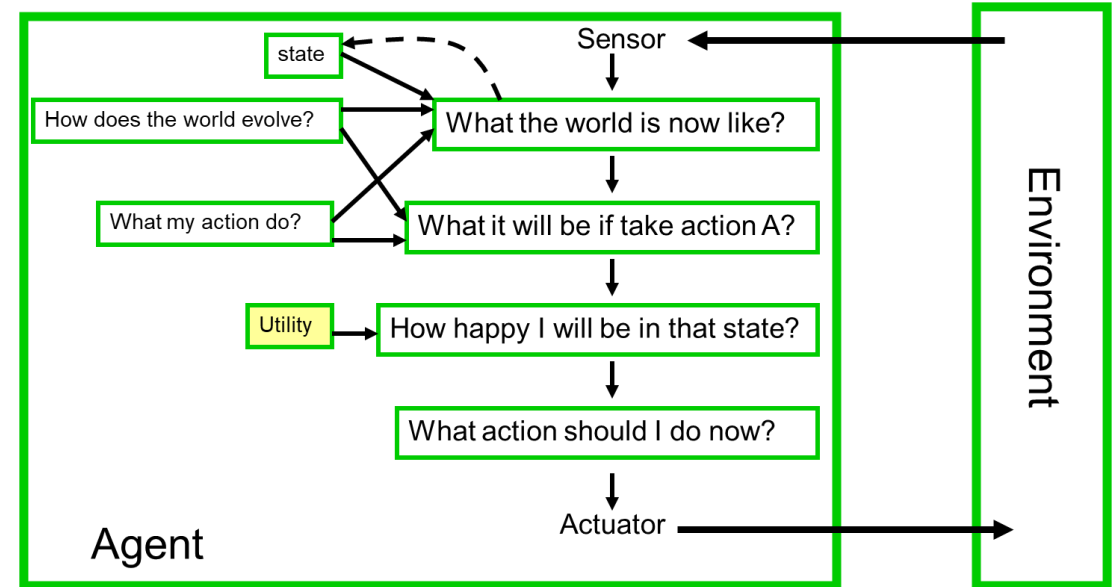
Agent Type	Description	Key Characteristics	Strengths	Limitations
Goal-Based Agents	Make decisions based on achieving specific goals rather than just reacting to conditions.	- Uses goal information to decide actions. - May include search and planning techniques.	- More flexible than reflex-based agents. - Can adapt to dynamic environments.	- Requires more computation. - Can be inefficient if goal-setting is unclear.

Goal-based agents

- A **goal-based agent** makes decisions by evaluating possible actions in terms of achieving a specific goal rather than just reacting to current conditions. It requires **knowledge of the world** and a way to assess whether an action moves it closer to its goal.
 - Example: A pathfinding Agent that chooses the shortest route based on a destination.

Utility-Based Agents

- Can evaluate **how good** a goal state is
- Choose actions that **maximize expected utility**
- Needed when there are **conflicting goals** or **uncertainty**
- **Example:**
Choose path A (fast but risky) vs. path B (safe but slow) based on utility function.



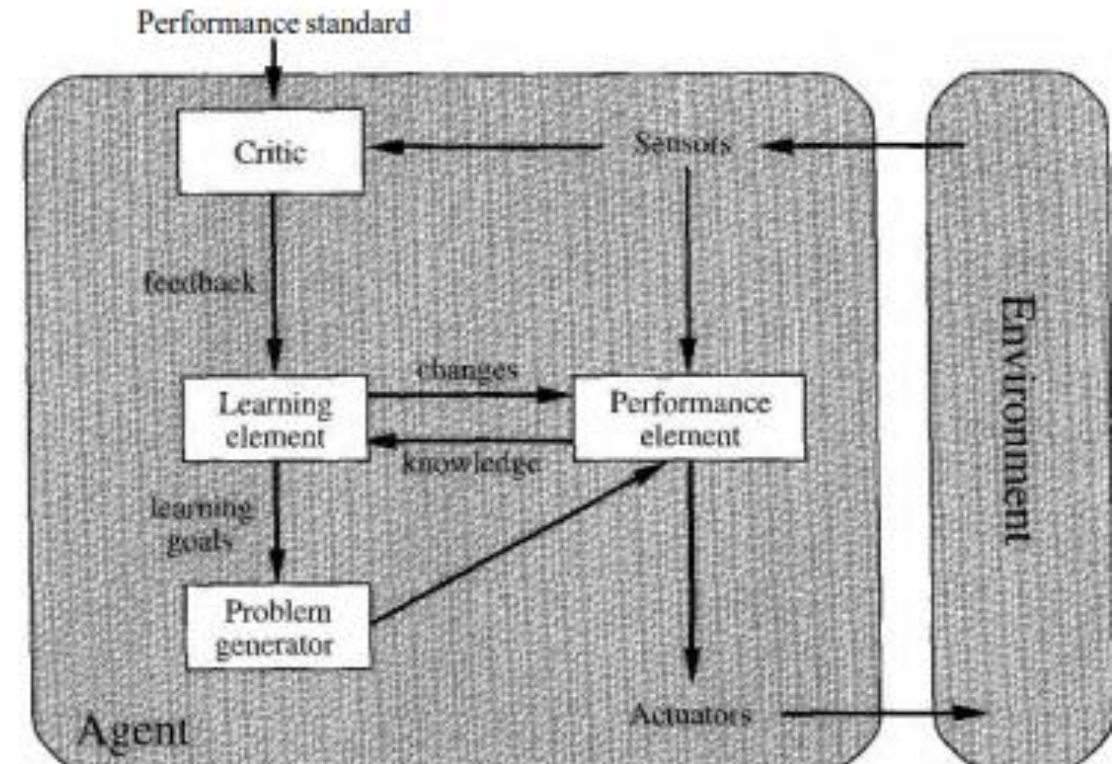
Agent Type	Description	Key Characteristics	Strengths	Limitations
Utility-Based Agents	Choose actions that maximize a utility function, aiming for the best possible outcome.	- Evaluates different options to maximize performance. - Uses utility measures for better decision-making.	- Provides optimal decision-making. - Handles trade-offs between different choices.	- Complex to implement. - Requires a well-defined utility function.

Utility-based agents

- A **utility-based agent** is a more advanced form of goal-based agents that evaluates possible actions **not just based on whether they achieve a goal, but on how "desirable" or "optimal" each outcome is**. These agents maximize an overall utility function, which assigns numerical values to different states to determine the best course of action.

Learning Agents

- Can improve performance over time
- Components:
 - Learning Element
 - Performance Element
 - Critic
 - Problem Generator



Learning Agents

Agent Type	Description	Key Characteristics	Strengths	Limitations
Learning Agents	Improves performance by learning from experience and feedback	- Updates behavior over time - Uses critic & performance modules - Explores new strategies via problem generator	- Adapts to unknown or changing environments - Learns from experience - Reduces need for full prior knowledge	- Requires sufficient training time - May perform poorly while learning - Design complexity (balancing exploration and exploitation)

A learning agent is one that can improve its performance over time by learning from its experiences in the environment.

It has components that allow it to **observe feedback, update its knowledge, and adapt its future behavior** accordingly.

- These agents are especially useful when **the environment is dynamic or unknown**, and when it's not feasible to pre-program all necessary knowledge.

Example of Different Types

Agent Type	Why / When Used?
Simple Reflex	Lane-keeping (e.g., if car drifts → steer slightly)
Model-Based Reflex	Use internal map to navigate a city block
Goal-Based	Plan route from point A to B while obeying traffic laws
Utility-Based	Decide between fastest vs. safest route depending on road/weather
Learning Agent	Improve decision-making over time by learning from traffic conditions and passenger feedback

Reading

- Chapter 2 of the Textbook